Report: project on absolute stability regions for ERK, AB and AM methods

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Chapter 1

Description of the problem and objectives

The goal of the chosen project was to construct a Matlab tool allowing to display the contours of absolute stability regions for three specific numerical methods for the solution of initial value problems (IVP): Explicit Runge-Kutta (ERK), Adams-Bashforth (AB) and Adams-Moulton (AM). The tool should be able to draw A-stability regions for those methods with order between 1 and 5, and also to eventually show all the regions for each family (ERK, AB, AM) separately.
We are going to discuss the problem and to describe how the tool works inside this report.

1.1 A-stability regions and A-stable methods

The notion of A-stability is given in order to describe the behaviour of a method on stiff problems. It can be analyzed by applying the method to the **test problem**

$$y'(t) = \lambda y(t) \ ; \ y(0) = 1$$

where $\lambda \in \mathbb{C}$.
The true solution of this problem is known to be $y(t) = y_0 e^{\lambda t}$. If $\Re\lambda < 0$ then the solution is such that $|y(t_1)| < |y(t_2)|$ if $t_1 > t_2$. Hence, given a method $y_n = y_{n-1} + h\Phi(t_{n-1}, y_{n-1}, t_n, y_n, \lambda y, h)$ we want in particular $|y_n| < |y_{n-1}| \ \forall n$.

Definition 1 *(absolute stability region)*
We define ***absolute stability region*** *the set*

$$\mathcal{H}_a = \{h\lambda \in \mathbb{C}^- \ \textit{such that} \ |y_n| < |y_{n-1}| \ \ \forall n\}$$

.

Definition 2 *(absolute stable method)*
A numerical method for IVP problems is said to be Absolute stable (or A-stable) if $\mathbb{C}^- \subseteq \mathcal{H}_a$, where $\mathcal{H}_a$ is the Absolute stability region of the method.

Chapter 2

Numerical methods and utilized software tools

2.1 A-stability regions for ERK:

For $p \leq 4$ and $f(t, y) = \lambda y$ the ERK formulas can always be written as:

$$y_n = y_{n-1}\left(1 + (h\lambda) + \frac{(h\lambda)^2}{2!} + \cdots + \frac{(h\lambda)^p}{p!}\right) \tag{2.1}$$

$\Rightarrow |y_n| < |y_{n-1}| \Leftrightarrow \left|\left(1 + (h\lambda) + \frac{(h\lambda)^2}{2!} + \cdots + \frac{(h\lambda)^p}{p!}\right)\right| < 1.$

Hence, the absolute stability region is given by

$$H_a = \left\{h\lambda \in \mathbb{C}^- \text{ s.t. } \left|\left(1 + (h\lambda) + \frac{(h\lambda)^2}{2!} + \cdots + \frac{(h\lambda)^p}{p!}\right)\right| < 1\right\} \tag{2.2}$$

We plot the boundary of this sets for $p = 1, ..., 4$ on Matlab using the *contour* command which allows the user to draw countour lines.
This is an example for $p = 2$:

```
[X,Y] = meshgrid(-4:0.01:4,-4:0.01:4);
          Mu = X+i*Y;
          R = 1 + Mu + .5*Mu.^2 ;
          Rhat = abs(R);
          contour(app.UIAxes2,X,Y,Rhat,[1,1],'r');
```

For $p = 5$, (2.1) doesn't hold. In particular we cannot write a fifth order method using only 5 stages. In general, for a ERK5 formula with 6 stages, the stability function will be:

$$R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24} + \frac{z^5}{120} + Cz^6$$

with C depending on the chosen butcher matrix. [1]

We chose the butcher:

$$
\begin{array}{c|cccccc}
0 & & & & & & \\
\frac{1}{4} & \frac{1}{4} & & & & & \\
\frac{1}{4} & \frac{1}{8} & \frac{1}{8} & & & & \\
\frac{1}{2} & 0 & 0 & \frac{1}{2} & & & \\
\frac{3}{4} & \frac{3}{16} & \frac{-3}{8} & \frac{3}{8} & \frac{9}{16} & & \\
1 & \frac{-3}{7} & \frac{8}{7} & \frac{6}{7} & \frac{-12}{7} & \frac{8}{7} & \\
\hline
 & \frac{7}{90} & 0 & \frac{32}{90} & \frac{12}{90} & \frac{32}{90} & \frac{7}{90}
\end{array}
$$

and we obtained $C = \frac{1}{1280}$.

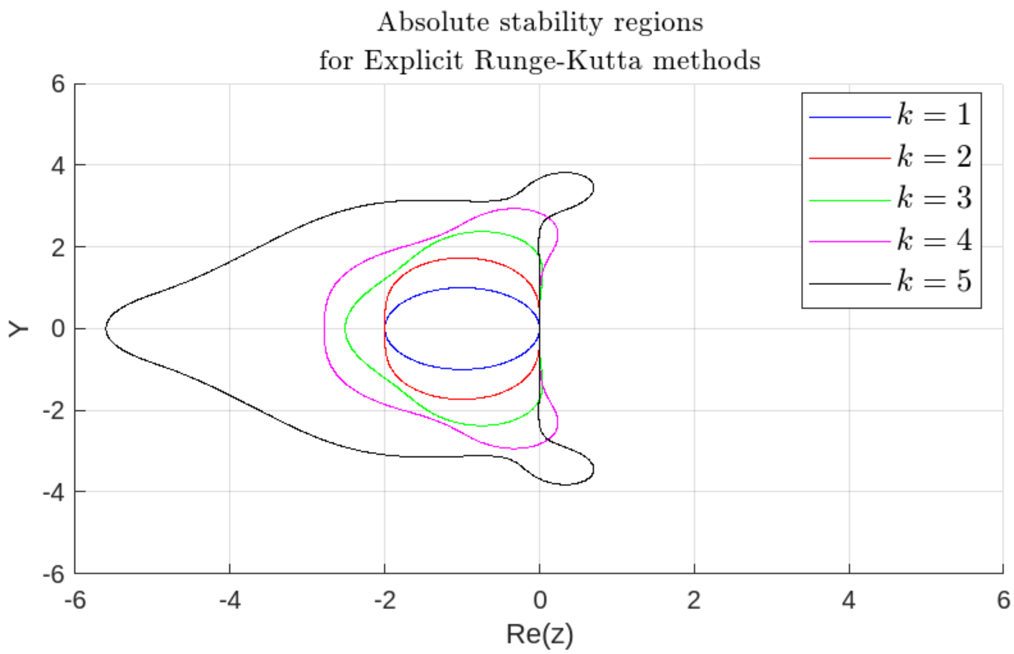


Figure 2.1: Absolute stability regions obtained for ERK formulas, order from 1 to 5

2.2 A-stability for Linear multistep methods (LMM): AB and AM

The stability regions for linear multistep methods are computed taking advantage of the finite difference equations theory.

Definition 3 *A finite difference equation $\Phi(\xi) = \sum_{i=0}^{k} a_j \xi^{k-j}$ is defined stable if its roots ξ_i are such that $|\xi_i| \leq 1\ \forall i = 1, ..., k$ and $|\xi_i| \leq 1 \Rightarrow \xi_i$ is a simple root. Moreover, it's said strongly stable if $|xi| < 1\ \forall i$.*

Considering now the LMM method

$$\sum_{j=0}^{k} \alpha_j y_{n-j} = h \sum_{j=0}^{k} \beta_j f_{n-j} \tag{2.3}$$

and applying it to the test problem $y' = \lambda y$ with initial condition $y(0) = 1$ we can rewrite (2.3) as

$$\sum_{j=0}^{k} (\alpha_j - h\lambda\beta_j) r^j = 0 \tag{2.4}$$

where $r_j = y_{n-j}$.
Defining

$$\rho(r) = \sum_{j=0}^{k} \alpha_j r^j \quad \text{and} \quad \sigma(r) = \sum_{j=0}^{k} \beta_j r^j \tag{2.5}$$

we can write the characteristic polynomial for the coefficients of the LMM as

$$\Phi(r) = \rho(r) - h\lambda\sigma(r) \tag{2.6}$$

It can be shown that the absolute stability region for this methods is

$$\mathcal{H}_a = \{\lambda h \in \mathbb{C}^- \text{ s.t. } |r_i| < 1\ i = 1, ..., k\}$$

with r_i roots of Φ [2].
Let $\partial\mathcal{H}_a$ be the boundary of the stability region, then $z \in \partial\mathcal{H}_a \Leftrightarrow$ corresponds to a root r with $|r| = 1$ then. Writing $r = e^{i\theta}$ we have

$$z = \frac{\rho(e^{i\theta})}{\sigma(e^{i\theta})} \tag{2.7}$$

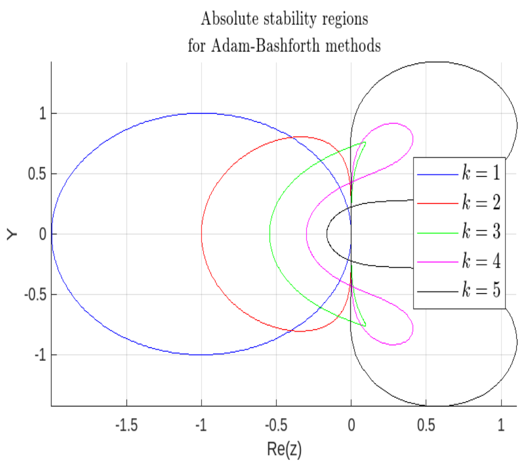
We can use (2.7) to easily plot the absolute stability regions for Adams-Bashfort for $p > 1$ and Adams-Moulton method for $p > 2$. Observe that in these cases $\rho(r) = r^n - r^{n-1}$ as we have $\alpha_0 = 1$, $\alpha_1 = -1$ and $\alpha_i = 0$ for any other i.
We used the following Matlab script, that has been generalized from one in the course notes

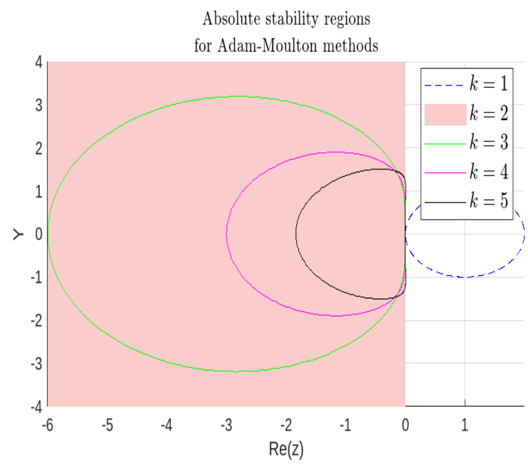
```
 theta=linspace(0,2*pi,500);
u_imag=sqrt(-1);
P=[];
for i = 1:numel(theta)
    xi=exp(u_imag*theta(i));
    yi=xi.^n *(xi-1)./s(xi);
   P=[P;[real(yi),imag(yi)]];
end

plot(P(:,1),P(:,2),'b');
grid on;
```

where $s(x)$ is a function computing the polynomial σ in x.

As AB1 and AM1 corresponds respectively to Explicit and Backward Euler, we had just to plot two circumferences of radius 1 (centered in -1 for AB1, in 1 for AM1), while for AM2 we used the command *xregion* to colour the half-plane of the negative abscissae, as AM2 is equivalent to the Trapezoid method.





2.3 Our Matlab GUI Toll

Using Matlab GUI Tool, we designed a simple application that allows the user to display the A-stability regions basically in two ways:

- Select the desired method from a Drop-Down list and plot all its A-stability regions for order k from 1 to 5. The user can do it in the first Tab, called "Absolute Stability Regions, varying method".
- Select the prescribed order from a drop-down list and the methods we want to compare from a check-list. The user can do it in the second Tab, called "Absolute Stability Regions, varying p".

In the case of unbounded regions we decided for AM1 to plot the countour line using a dashed line, meaning we are taking the exterior of the curve and to simply colouring the area of the region, for AM2. We decided not to let the user compare different methods for multiple values of k for two main reasons:

1 The "hidden" goal of comparing methods is to see which is the more stable having fixed an order of accuracy. We are not really interested in comparing stability regions for methods that have very different behaviours concerning convergence.

2 Maintaining the plots readable. As this regions are mostly bounded and next to the origin we can see that they intersect a lot: better not to plot too many of them.

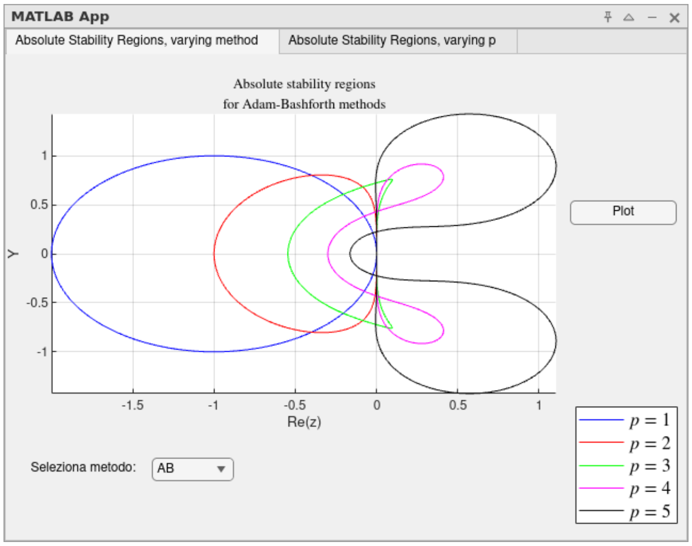


Figure 2.3: Screenshot from the Matlab GUI tool: A-stability regions for Adams-Bashforth formulas

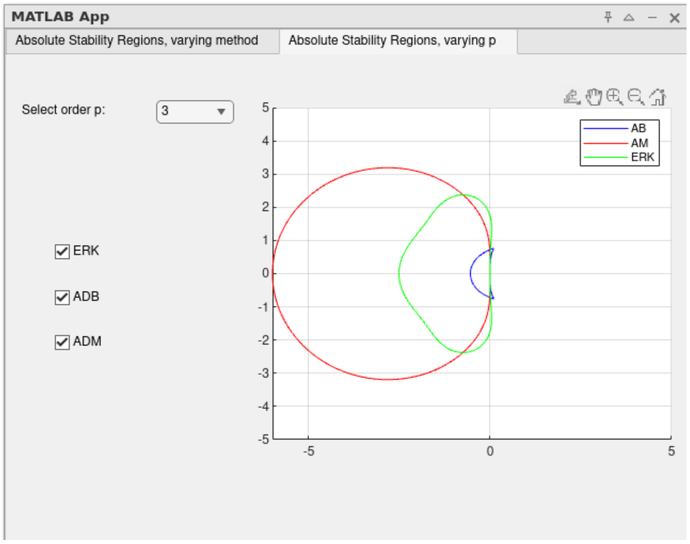


Figure 2.4: Screenshot from the Matlab GUI tool: comparing A-stability regions for ERK, AB and AM of order 3.

Chapter 3

Discussion and analysis of the results

We were able to plot all of the requested Absolute stability regions.
The results reflect what we expected for different reasons.
We expected all the Runge-Kutta A-stability regions to be bounded: this is due to the fact all the stability regions $R(z)$ for ERK methods can be written as polynomials, then

$$\lim_{|z|\to\infty} |R(z)| = \infty$$

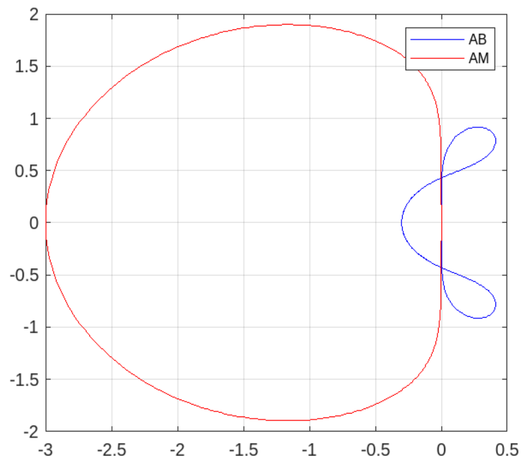
More in general, the only unbounded regions obtained are the ones relative to Adams-Moulton with order 1 and 2 which corresponds respectively to Backward Euler and the Trapezoid method, which are A-stable. This results in line with the Dahlquist barrier, which can be stated as follows:

"No LMM method of order greater than 2 is A-stable."

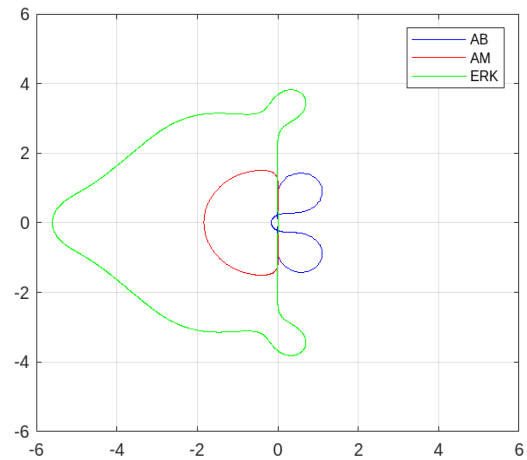
With regard to comparisons between methods we report the following observations:

- Even when A-stability regions of AB and AM are both bounded (i.e. $p > 2$) we note that those of the implicit method are much larger for any p.
- ERK regions also are greater than AB regions, except the case $p = 1$ when both the methods are equivalent to Explicit Euler.
- ERK regions tend to become larger as p increases, while the LMM methods ones tend to decrease in size, both for Adams-Bashfort and for Adams-Moulton.

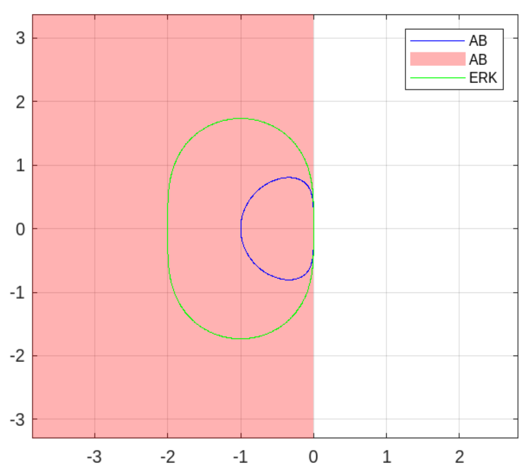
We attach some screenshot from the GUI tool:



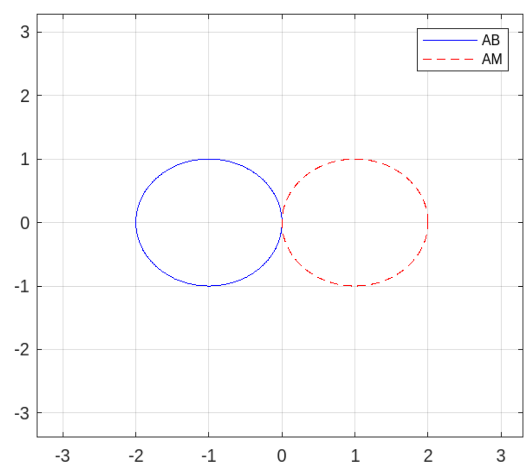
(a) Comparison between order 4 AM and AB.



(b) Comparison between order 5 ERK, AB and AM



(a) Comparison between order 2 AB, ERK and AM (trapezoid)



(b) Comparison between order 1 AB (Explicit Euler) and AM (Backward Euler).

Figure 3.2: Recall: The dashed line means we are taking the exterior of the curve

Chapter 4

Conclusions

Our very simple tool enables the user to address two important problems concerning Absolute stability- even if in a quite restrictive context as we are dealing with just three families of methods and considering only orders $p \leq 5$-:
verify how, given a method, its stability regions evolve for p increasing;
given a order p, compare different methods so that we can check which is the most stable.
We can do that graphically and using only a small number of commands.
It would be nice to add to this tool other methods, like BDF or PECE, or giving the possibility to choose different butcher matrixes for ERK with order greater than four.
With regard to the obtained A-stability regions, we could conclude that Adams-Bashforth method results no doubt the least stable, as its regions are always much smaller than the others.
About the other two methods, we could say that Adams-Moulton appears to be more stable for $1 \leq p \leq 3$, Explicit Runge-Kutta for $p = 5$. With regard to the case $p = 4$, we can say that the stability of the two methods does not differ significantly.
We have to remark that we cannot establish which method should be preferable by just referring to the size of its A-stability region, since other relevant factors, such as the number of function evaluations, need to be considered. However, as far as explicit methods is concerned: since ERK and AB methods use approximately the same number of function evaluations (both p per step if $p \leq 4$, ERK uses 6 per step if $p = 5$, AB 5) but the first of them appears always more stable, we should usually prefer it. More delicate is taking into account Adams Moulton, since the number of function evaluations can vary a lot for implicit methods, depending on the nature of the particular IVP problem.

Bibliography

[1] John Charles Butcher. *Numerical methods for ordinary differential equations*. John Wiley & Sons, 2016.

[2] Randall J LeVeque. *Finite difference methods for ordinary and partial differential equations: steady-state and time-dependent problems*. SIAM, 2007.